

Задание 20 — модули ядра (часть 1)

Вариант №1: базовый модуль ядра

1) Исходный код:

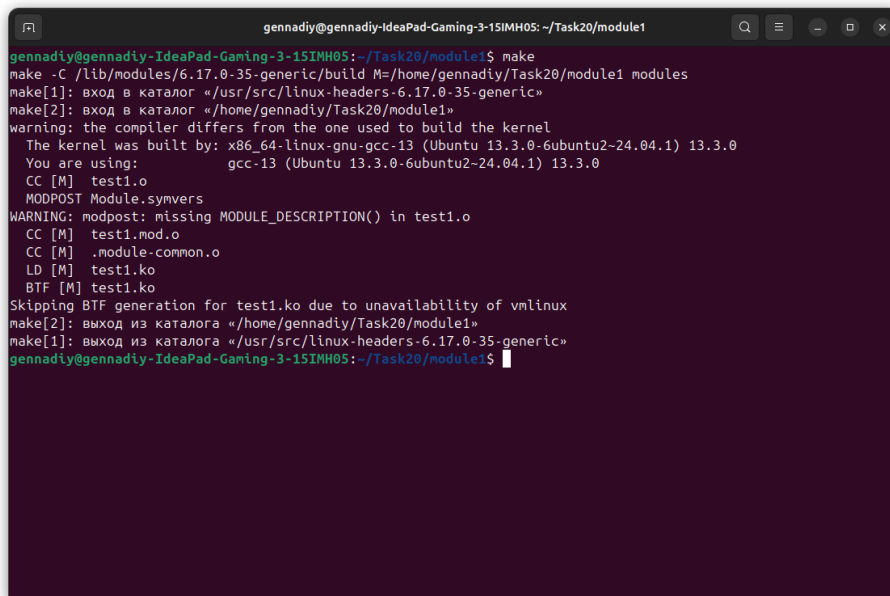
```
#include <linux/module.h>
#include <linux/kernel.h>

int init_module(void)
{
    pr_info("My test module loaded!!!!\n");

    return 0;
}

void cleanup_module(void)
{
    pr_info("My test module unloaded!!!!\n");
}

MODULE_LICENSE("GPL");
```



```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/Task20/module1
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module1$ make
make -C /lib/modules/6.17.0-35-generic/build M=/home/gennadiy/Task20/module1 modules
make[1]: вход в каталог «/usr/src/linux-headers-6.17.0-35-generic»
make[2]: вход в каталог «/home/gennadiy/Task20/module1»
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
You are using: gcc-13 (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
CC [M] test1.o
MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in test1.o
CC [M] test1.mod.o
CC [M] .module-common.o
LD [M] test1.ko
BTF [M] test1.ko
Skipping BTF generation for test1.ko due to unavailability of vmlinux
make[2]: выход из каталога «/home/gennadiy/Task20/module1»
make[1]: выход из каталога «/usr/src/linux-headers-6.17.0-35-generic»
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module1$
```

Рисунок 1 — результаты сборки

2) Используемые команды:

`sudo dmesg | tail -n 10` — просмотр системного лога

`sudo insmod test1.ko` — загрузка модуля в пространство ядра

`sudo rmmod test1` — выгрузка модуля из ядра

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/Task20/module1
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module1$ sudo insmod test1.ko
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module1$ sudo dmesg | tail -n 10
[14899.737171] audit: type=1400 audit(1781721759.802:116345): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.dirs" pid=903669 comm="sed" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[14899.737533] audit: type=1400 audit(1781721759.802:116346): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.dirs" pid=903670 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[14899.739375] audit: type=1400 audit(1781721759.804:116347): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.locale" pid=903671 comm="cp" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[14899.739705] audit: type=1400 audit(1781721759.804:116348): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.locale" pid=903672 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[14899.741774] audit: type=1400 audit(1781721759.806:116349): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/etc/xdg/user-dirs.conf" pid=903673 comm="xdg-user-dirs-u" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[14899.769035] audit: type=1400 audit(1781721759.834:116350): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/usr/share/icons/" pid=903611 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[14899.769057] audit: type=1400 audit(1781721759.834:116351): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/var/lib/snapd/desktop/icons/" pid=903611 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[14899.820831] audit: type=1400 audit(1781721759.885:116352): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/usr/share/glib-2.0/schemas/" pid=903747 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[14899.822852] audit: type=1400 audit(1781721759.887:116353): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/usr/share/glib-2.0/schemas/" pid=903749 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[14900.965846] My test module loaded!!!!
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module1$
```

Рисунок 2 — системный лог при загрузке

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/Task20/module1
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module1$ sudo rmmod test1
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module1$ sudo dmesg | tail -n 10
[15172.232690] audit: type=1400 audit(1781722032.298:118481): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.dirs" pid=920107 comm="sed" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[15172.233113] audit: type=1400 audit(1781722032.299:118482): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.dirs" pid=920108 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[15172.234908] audit: type=1400 audit(1781722032.301:118483): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.locale" pid=920109 comm="cp" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[15172.235296] audit: type=1400 audit(1781722032.301:118484): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.locale" pid=920110 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[15172.248521] audit: type=1400 audit(1781722032.314:118485): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/usr/share/icons/" pid=920050 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[15172.248541] audit: type=1400 audit(1781722032.314:118486): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/var/lib/snapd/desktop/icons/" pid=920050 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[15172.271230] audit: type=1400 audit(1781722032.333:118487): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/usr/share/glib-2.0/schemas/" pid=920168 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[15172.273104] audit: type=1400 audit(1781722032.338:118488): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/usr/share/glib-2.0/schemas/" pid=920170 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[15172.306683] audit: type=1400 audit(1781722032.372:118489): apparmor="DENIED" operation="open" class="file"
profile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/etc/fonts/snap-override/" pid=920186 comm="5B70616E676F5D20666F6E74636F6E" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[15173.741127] My test module unloaded!!!!
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module1$
```

Рисунок 3 — системный лог при выгрузке

Вариант №2: модуль с файлом устройства

1) Исходный код:

```
#include <linux/module.h>
#include <linux/kernel.h>
#include <linux/fs.h>
#include <linux/rwlock.h>
#include <linux/uaccess.h>
```

```
static int major = 0;
static rwlock_t lock;
static char test_string[15] = "Hello!\n\0";
```

```
ssize_t test_read(struct file *fd, char __user *buff, size_t size, loff_t *off)
```

```

{
    size_t rc;

    read_lock(&lock);
    rc = simple_read_from_buffer(buff, size, off, test_string, 15);
    read_unlock(&lock);

    return rc;
}

ssize_t test_write(struct file *fd, const char __user *buff, size_t size, loff_t *off)
{
    size_t rc = 0;
    if (size > 15)
        return -EINVAL;
    // Отредактировано
    write_lock(&lock);
    if (__copy_from_user(test_string, buff, size) == 0) {
        rc = size;
    } else {
        rc = -EFAULT;
    }
    write_unlock(&lock);
    // Отредактировано
    return rc;
}

static struct file_operations fops = {
    .owner = THIS_MODULE,
    .read = test_read,
    .write = test_write
};

int init_module(void)
{
    pr_info("Test module is loaded.\n");
    rwlock_init(&lock);
    major = register_chrdev(major, "test2", &fops);

    if (major < 0)
        return major;
    pr_info("Major number is %d.\n", major);

    return 0;
}

void cleanup_module(void)
{
    unregister_chrdev(major, "test2");
}

MODULE_LICENSE("GPL");

```

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/Task20/module2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$ make
make -C /lib/modules/6.17.0-35-generic/build M=/home/gennadiy/Task20/module2 modules
make[1]: вход в каталог «/usr/src/linux-headers-6.17.0-35-generic»
make[2]: вход в каталог «/home/gennadiy/Task20/module2»
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
You are using: gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
CC [M] test2.o
test2.c:11:9: warning: no previous prototype for 'test_read' [-Wmissing-prototypes]
 11 | ssize_t test_read(struct file *fd, char __user *buff, size_t size, loff_t *off)
    |
test2.c:22:9: warning: no previous prototype for 'test_write' [-Wmissing-prototypes]
 22 | ssize_t test_write(struct file *fd, const char __user *buff, size_t size, loff_t *off)
    |
MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in test2.o
CC [M] test2.mod.o
CC [M] .module-common.o
LD [M] test2.ko
BTF [M] test2.ko
Skipping BTF generation for test2.ko due to unavailability of vmlinux
make[2]: выход из каталога «/home/gennadiy/Task20/module2»
make[1]: выход из каталога «/usr/src/linux-headers-6.17.0-35-generic»
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$
```

Рисунок 4 — результаты сборки

2) Используемые команды:

`sudo insmod test2.ko` - загружаем модуль в ядро

`sudo dmesg | tail -n 10` - смотрим логи ядра, чтобы узнать Major-номер устройства

`sudo mknod /dev/test2 c 504 0` - создаем файл устройства

`sudo chmod 666 /dev/test2` — даем права на чтение и запись всем пользователям

`head -n 1 /dev/test2` - читаем файл устройства

`echo "Hi" > /dev/test2` - записываем новую строку в файл устройства

`sudo rm /dev/test2` - удаляем файл устройства

`sudo rmmod test2` - выгружаем модуль

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/Task20/module2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$ sudo rmmod test2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$ sudo insmod test2.ko
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$ sudo dmesg | tail -n 10
[23070.469773] audit: type=1400 audit(1781729930.558:180218): apparmor="DENIED" operation="open" class="file" prof
ile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.locale" pid=
1398911 comm="cp" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[23070.470082] audit: type=1400 audit(1781729930.558:180219): apparmor="DENIED" operation="open" class="file" prof
ile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/home/gennadiy/.config/user-dirs.locale" pid=
1398912 comm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[23070.471729] audit: type=1400 audit(1781729930.560:180220): apparmor="DENIED" operation="open" class="file" prof
ile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/etc/xdg/user-dirs.conf" pid=1398913 comm="xd
g-user-dirs-u" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[23070.500550] audit: type=1400 audit(1781729930.589:180221): apparmor="DENIED" operation="open" class="file" prof
ile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/usr/share/icons/" pid=1398852 comm="desktop-
launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[23070.500588] audit: type=1400 audit(1781729930.589:180222): apparmor="DENIED" operation="open" class="file" prof
ile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/var/lib/snapd/desktop/icons/" pid=1398852 co
mm="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[23070.521289] audit: type=1400 audit(1781729930.609:180223): apparmor="DENIED" operation="open" class="file" prof
ile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/usr/share/glib-2.0/schemas/" pid=1398987 com
m="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[23070.523064] audit: type=1400 audit(1781729930.611:180224): apparmor="DENIED" operation="open" class="file" prof
ile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/usr/share/glib-2.0/schemas/" pid=1398989 com
m="desktop-launch" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[23070.553369] audit: type=1400 audit(1781729930.641:180225): apparmor="DENIED" operation="open" class="file" prof
ile="snap.snapd-desktop-integration.snapd-desktop-integration" name="/etc/fonts/snap-override/" pid=1399005 comm=5
B70616E676F5D20666F6E74636F6E requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[23071.555005] Test module is loaded.
[23071.555009] Major number is 504.
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$
```

Рисунок 4 — системный лог при загрузке

```
B70616E676F5D20666F6E74636F6E requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[23071.555005] Test module is loaded.
[23071.555009] Major number is 504.
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$ sudo mknod /dev/test2 c 504 0
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$ ls -la /dev/test3
ls: невозможно получить доступ к '/dev/test3': Нет такого файла или каталога
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$ ls -la /dev/test2
crw-r--r-- 1 root root 504, 0 июн 18 04:00 /dev/test2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$ head -n 1 /dev/test2
Hello!
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$ sudo chmod 666 /dev/test2
```

Рисунок 5 — создание файла устройства и настройка прав доступа

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$ echo "Hi" > /dev/test2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$ head -n 1 /dev/test2
Hi
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/Task20/module2$
```

Рисунок 6 — записываем новую строку и читаем обратно